

V SHREYAS

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EDUCATION

Indian Institute of Technology, Roorkee

July 2012 - June 2017

Dual Degree M.Tech/B.Tech

Electronics and Comm Engineering with Specialisation in Wireless Comm

GPA: 7.86/10

Stanford AI Professional Program

May 2024 - Aug 2026

Foundational Models, Fine-tuning, Deep Reinforcement Learning

SUMMARY

Quant and data science professional with over 9 years in model risk management, data science, and quantitative analytics for global financial institutions. Supported \$3B+ AUM, led teams up to 4 members, and developed high-impact analytics platforms. CFA Level III passed with a background of research in AI & ML.

WORK EXPERIENCE

Capital One, Bengaluru, India

April 2023 - Present

Data Science Manager, Trading Model Risk Management

- One of the leads in the modernization of Model Risk Management with the help of AI tools
- Owned and managed end-to-end model risk governance activities of the Balance Sheet Management and VaR (Value-at-Risk) models within trading risk, as well as the Counterparty PD & LGD models. Designated Near Model Risk Officer (NMRO), independently assessing model risks for Trading division. Managed 3 quant associates and 1 intern during this stint.
- Improved the model risk monitoring framework across SIMM and VaR model families, reducing model failures significantly through stronger validation logic, benchmark design, and framework enhancements.
- Enhanced the VaR backtesting framework by introducing analytics to detect p-value drift in PL distributions, improving early warning capability for model performance deterioration.
- Built benchmark models and conducted scenario analyses to challenge model assumptions and surface weaknesses in risk model design, calibration, and sensitivity to stressed conditions.
- Developed SQL-based data pipelines to generate and standardize reusable business scenarios across multiple asset classes within the Derivative Analytics Platform (DAP), improving consistency and reusability of scenario testing.
- Partnered with cross-functional stakeholders (risk, modeling, and business teams) to translate model risk review needs into scalable analytics workflows and reusable tooling.
- Initiated AI Foundations program, training 6 senior quants and growing rapidly across teams. Led problem formulation for the first nationwide hackathon organized by Capital One. [Capital One Launchpad](#)

Franklin Templeton Investments, Hyderabad, India

April 2020 - March 2023

Data Scientist, Floating Rate Group

- Supported \$3 billion in AUM and established a data science effort for the Floating Rate Group. Worked on CLO optimization for return maximization within risk constraints using heuristic methods.
- Contributed to the development and implementation of a recommendation system, which included a recommendation search platform and quarterly attribution of analyst recommendations.
- Contributed to an XGBoost-based HY/IG bond selection model for factor-based investing, supporting portfolio decision-making with data-driven ranking signals. Utilized the same framework and extended the model for leveraged loan selection.
- Led implementation of an ESG database and developed a structured financials engine from unstructured data using AWS Textract.

Capital One, Bengaluru, India

June 2017 - March 2020

Senior Quantitative Analyst

- Worked in assessment of the market, counterparty and operational risk models as a part of the Model Risk Office. Been part of 3 successful regulatory exams within the company.
- Independently built and developed a risk assessment and valuation tool for commodity derivatives in Capital One portfolio.
- Led validation efforts of commodity risk models, including review of new product releases like power derivatives and commodity futures. Recognized by multiple stakeholders for work in the market and commercial risk space.

Capital One, Bengaluru, India

May 2016 - July 2016

Intern

- Worked with the Enterprise Data Management team to implement an in-house facility of data-on-tap for analysis. This reduced the team's review time for metadata mapping and led to better interpretability of data sources.

Moonfrog Labs, Bengaluru, India

May 2015 - July 2015

Intern

- Reduced network latency by developing a cross-platform analytics library with efficient data compression. This helped improve user experience.

AWARDS AND RECOGNITION

- Awarded MHRD (Ministry of Human Resources) scholarship for M.Tech in 2017.
- CFA Level III passed, 2024
- Received Capital One India's highest award (Icon of Excellence, 2025) for exemplary management and execution
- Stanford AI Professional Programme Certifications -
 - [XCS234 - Reinforcement Learning](#)
 - [XCS 224U - Natural Language Understanding with Deep Learning](#)
 - [Deep Reinforcement Learning](#)
- Kishore Vaigyanik Protsahan Yojana (KVPY) scholar, 2011

SELECTED DATA SCIENCE PROJECTS

- Implemented Deep Q-Network based architectures for Atari Pong, Monte-Carlo based REINFORCE for multiple games, and Contextual Bandit for personalized recommendations
- Fine-tuned ELECTRA-small for sentiment analysis and BERT for compositional generalization. Implemented a simplified RAG system for question answering using DSPy

PUBLICATIONS

SHREYAS, V., AND VINOD, P. A Deep Learning Architecture for Brain Tumor Segmentation in MRI Images. In *IEEE 19th International Workshop on Multimedia Signal Processing (MMSP), London, UK* (2017)

TECHNICAL STRENGTHS

- **Programming Languages:** Python, R, SQL, C++, C#, JavaScript
- **ML / Stats:** Prediction Models (Boosted Trees, SVM, Linear models), Model Evaluation and Risk Management, Optimization, Pandas, Scikit-learn
- **Deep Learning / NLP:** PyTorch, TensorFlow, DSPy, Reinforcement Learning, Convolutional Neural Networks
- **Quant / Finance Analytics:** derivatives valuation, sensitivities/Greeks, scenario analysis, Value-at-Risk (VaR), Initial Margin Model, Potential Future Exposure, Credit Value Adjustment (xVA)
- **Cloud Tools:** AWS AI Services, Azure Cognitive Services, Git, Claude CLI
- **Technical Writing**